SMART A PLUS

UNIT 5

ELECTRIC ENERGY : CONSUMPTION AND CONSERVATION

Answer the following questions. Each question carries 1 score

1. Carefully read the following statement and reason and then chose the correct option

Statement: Heat generated in a conductor increases when length increases.

Reason: Resistivity of a material increases when length of the material increases

A. Both statement and reason are true, reason is the correct explanation of the statement
B. Statement is true and the reason is false, reason is the correct explanation of the statement
C. Both statement and reason are false
D. Statement is true and the reason is false, reason does not explain the statement

2. Two conducting wires of the same material and of equal lengths and equal diameters are first connected in series and then parallel in a circuit across the same potential difference. The ratio of heat produced in series and parallel combinations would be

A. 1:1
B. 1:2
C. 4:1
D. 1:4

3. 'Heating elements of electrical appliances are made of nichrome'.

Select the correct reason(s) of the above statement from the following

i. Nichrome has high resistivity and high melting point.
ii. Nichrome has high resistivity and low melting point.
iii. Nichrome has an ability to provide heat energy for a long time in a red hot state.
iv. Nichrome has low oxidation resistance

i and ii are correct
iii and iv are correct
i and iii are correct
ii and iii are correct

4. In a house, an electric iron of 600 W works for 10 minutes and a fan of 60 W works for 10 hours. Chose the correct option from the following based on their energy consumption

A. energy consumed by the electric iron is 10 times higher than the fan
B. energy consumed by the electric iron is 6 times higher than the fan
C. Iron box and fan consumes same amount of energy
D. energy consumed by the electric fan is 10 times higher than the iron box

5. Analyze the following assertion and reason and chose the correct option

Assertion (A): A low-power appliance can consume more energy than a high-power appliance. **Reason (R):** Energy consumed depends on both power and time of usage.

A. A and R are true, and R explains A
B. A and R are true, but R does not explain A
C. A is true, R is false
D. A is false, R is true

Answer the following questions. Each question carries 2 score

6. Two coils A and B are connected parallel to a battery. If the resistance of the coil A and B are 60 Ω and 100 Ω, respectively and the heat developed in the coil B is 420 J in 20 s, then how much heat (in J) is produced in the coil A in 60 s.

7. Solar panels are devices that convert solar energy into electric energy. Write down any four benefits of installing solar panels

8. An electric kettle of resistance 100 Ω produces 240000 J heat when operated for 10 min. Calculate the current flowing through the kettle.

9. Current flowing in an electric heater decreases from 4 A to 2 A, when its internal resistance is increased from 20 Ω to 40 Ω. Calculate the change in the amount of heat emitted when it is used for 10 minutes. Heat generated in the heater decreases or increases? Justify your answer

10. When a resistance wire in a heating coil is replaced with another wire of the same material and length, but with twice the thickness, how the heat produced in the coil changes? (Assuming both are connected to same potential difference)

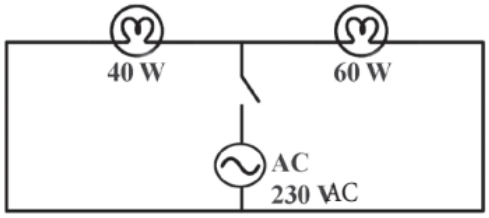
11. Can two heating coils of same material produce different heat? Explain

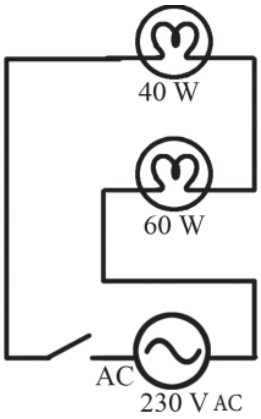
12. The part used to produce heat energy in heating appliances is the heating element.

a) Which material is used for making heating coil

b) What are the characteristics of the material suitable for the heating element?

13. Two bulbs, 230 V, 40 W and 230 V, 60 W are arranged as shown in the figures. If 230 V is applied, which bulb will be brighter in each circuit? Explain the reason.





Answer the following questions. Each question carries 3 score

14. An electric bulb is rated as 230V, 100W

a) What is its power when it operated at 115 V?

b) Can we operate this device at 460 V? Justify your answer.

15 Electricity bill can be reduced by lowering electricity consumption.

a) In a house, two 60 W fans work for 8 hours per day. If these fans are replaced with BLDC fans of power 25 W, how the cost of electricity will reduces in a month due to this replacement. (cost of electricity is ₹ 5.00 per unit)

b) Write any two methods to reduce electricity consumption in our home

16. Heating appliances uses heating coils

(a) If the heating coil of a heater is cut to reduce its length by 20% then effect on its power will be (Increase by 20%, decrease by 25%, Increase by 25%, Decrease by 20%)

(b) The second coil heats up more quickly. Why?

17 Three resistance wires A, B and C of the same length and same area of cross section are taken.

a) A student says that the heat produced by A and B are equal, whereas the heat produced by A is greater than heat produced by C. Is this possible? Explain.

b) To achieve the same heat on A and C, what change should be made on A?

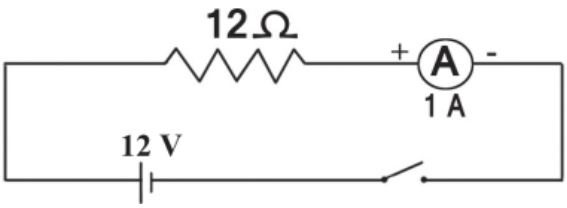
Answer the following questions. Each question carries 4 score

18. The quantity of greenhouse gases emitted, converted and expressed as equivalent to the measure of carbon dioxide, is carbon footprint.

a) 'Use of electric vehicles (EV) helps to reduce carbon footprint'. Do you agree with this statement? Justify your answer

b) Suggest any four methods to reduce carbon foot print?

19. Observe the circuits given below.



a) Calculate the quantity of heat produced in the resistor in 5 minutes.

b) State the law which helps you to calculate the heat developed in the resistor

c) If another one 12 Ω resistance is connected in parallel to the existing 12 Ω resistor, do the heat generated in the resistors increase or decrease. Justify your answer.

20. In a house, a 750 W electric iron operates for one hour, two 60 W fans for 8 hours, and five 10 W LED bulbs for 10 hours.

a) Calculate in unit the energy consumption per month in that house.

b) Calculate the total electricity bill for one month, If the cost of electricity is ₹ 5.00 per unit (kWh).

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Answers

1. Statement is true and the reason is false, reason does not explain the statement

2. 1:4
3. i and iii are correct
4. Iron box and fan consumes same amount of energy
5. A and R are true, and R explains A
6. Voltage same on both coils

   $H = V^2t/R$

   $V^2 = HR/t$

   ie $H_1R_1/t_1 = H_2R_2/t_2$

   $H_2 = H_1R_1\ t_2/R_2t_1 = 420x100x60/\ 60x20$

   $= 2100$ J
7. Can produce electricity required for homes.

   Can reduce environmental pollution.

   Transmission and distribution losses are minimised as electricity is generated at the place of consumption

   Reducing global warming (reduced by reducing the production of major greenhouse gases like carbon dioxide, methane et)
8. $H = I^2 R t$

   $I^2 = H/ R t = 240000 / (100x10x60) = 4$

   $I = 2$ A
9. change in the amount of heat $H = I_1^2 R_1 t_1 - I_2^2 R_2 t_2$

   $= 4^2$ x 20x 10x60 - 2^2 x 40x 10x60 = 96000 J

   Heat decreases. current decreases
10. Resistance is reduced to one-fourth (1/4). $R\alpha 1/A$, $A\alpha r^2$.

    Then the heat generated will become four times greater, since the voltage (V) remains the same.
11. Yes, two heating coils made of the same material can produce different amounts of heat. While the material determines the resistivity, the heat produced depends on the resistance, which is determined by the physical dimensions (length and cross-sectional area) of the coil. heat produced in a material changes when the length or area of cross section changes.
12.

    a) nichrome

    b) High oxidation resistance

    Ability to provide heat energy for a long time in a red hot state.

High resistivity

high melting point

13. $P = V^2/R$. $R = V^2/P$.

In a series circuit, the 40 W bulb will be brighter. Since I is constant, the bulb with the higher resistance R (the 40 W bulb) dissipates more power and glows brighter. ($H = I^2Rt$)

In a parallel circuit, the 60 W bulb will be brighter. Since V is constant, the bulb with the lower resistance R (the 60 W bulb) dissipates more power and glows brighter. ($H = V^2t/R$)

14. (a) Here resistance of the bulb is constant. $P = V^2/R$. $R = V^2/P$.

$P_2 = P_1 \times (V_2/V_1)^2 = 100 \times (115/230)^2 = 25$ W

(b) No, the device cannot be safely operated at 460 V. The power dissipated would be 400 W, which is four times its rated power of 100 W. This excessive power would cause too much heat, immediately burning out or fusing the bulb's filament.

15. (a)

$$\text{Electric energy (in kilowatt hour)} = \frac{\text{Power in watt} \times \text{time in hour}}{1000}$$

Normal fan, E = 2x60x8 / 1000 = 0.96 kWh

BLDC fan, E = 2x25x8 / 1000 = 0.40 kWh

Difference in energy consumption per day = 0.96-0.40 = 0.56 kWh

Difference in amount of electricity per month = 0.56x 30x 5 = 84 rupeese

(b) Use energy efficient electrical appliances.

Turn off switches immediately after use.

Use LED tubes and bulbs.

Choose the size of the fan according to the size of the room.

Use BLDC fans.

16. The power of the heater will increase by 25%

The second coil heats up more quickly because it has a lower electrical resistance

17.

a) Yes. Possible, A and B are same material, Resistivity same. R also same since Area and length are same. A and C are differant material, resistivity of A> resistivity of C

b) length of A should decrease or Area of cross section of A should increase

18.

(a)Yes to some extent. Although EVs helps reduce carbon emission, the major part of electricity in our country is generated from fossil fuels.

(b) Reduce domestic energy consumption.

- Avoid wasting food.
- Use public transport.
- Reduce waste by utilising reusable products.
- Educate society about reducing carbon footprint.
- Perform energy consuming activities in an energy saving manner

19. $H = I^2 R t$

H = 1x12x5x60 = 3600 J

The quantity of heat produced in a current carrying conductor is directly proportional to the square of the current (I2) the resistance of the conductor (R) and the time (t) for which the current flows. This is Joule's law

Increases. because effective resistance decreases, current increases

20. (a)

$$\text{Electric energy (in kilowatt hour)} = \frac{\text{Power in watt} \times \text{time in hour}}{1000}$$

E electric iron = 750x 1 / 1000 = 0.75 kWh
E (2 fan) = 2x 60x 8 /1000 = 0.96 kWh
E (5 LED) = 5x 10x 10 / 1000 = 0.5 kWh
Energy consumption per day = .75 + .96+0.5 = 2.21 kWh
Energy consumption per day = 2.21 x30 = 66.3 kWh
Total electricity bill for one month = 66.3 x 5 = Rs 331.5